

Homework 1 - Set Notation, Counting, and Probability Axioms

Instructions: Please submit your homework as a single PDF on Gradescope. After uploading, please make sure each page is assigned to the correct question before submitting. Show your work clearly.

1. (8 points) A box contains 3 orange, 4 purple, 5 silver, and 6 black balls. A sample of 5 balls is chosen without replacement. How many **possible samples** of size 5 include all four colors? $360 + 540 + 720 + 900 = 2520$ samples
are color would have 2 balls:

Case #1: 2 orange balls $\binom{3}{2}\binom{4}{1}\binom{5}{1}\binom{6}{1} = 360$ Case #2: 2 purple balls $\binom{4}{2}\binom{3}{1}\binom{5}{1}\binom{6}{1} = 540$ Case #3: 2 silver balls $\binom{5}{2}\binom{3}{1}\binom{4}{1}\binom{6}{1} = 720$ Case #4: 2 black balls $\binom{6}{2}\binom{3}{1}\binom{4}{1}\binom{5}{1} = 900$

2. (5 points) A security badge number consists of two uppercase letters followed by three digits. The two letters may not be repeated, the first digit cannot be 0, and the remaining digits may be repeated. How many **different** badge numbers are possible?

Letter. Letter. Digits. Digits. Digits
26 x 25 x 9 x 10 x 10 = 585000 badge numbers
↓ ↓ ↓
exclude 0 repeat ok

3. (5 points) A committee of 4 people is selected from 6 faculty members and 5 graduate students. What is the **probability** that the committee contains exactly 2 faculty members and 2 graduate students?

Total $n = 11$
$$\frac{\binom{6}{2} \times \binom{5}{2}}{\binom{11}{4}} = \frac{150}{330} = \frac{5}{11}$$

4. (7 points) What is the **probability** that a hand of 5 cards chosen randomly and without replacement from a standard deck of 52 cards contains the ¹king of spades, exactly ²1 other king and ^{3,4}exactly 2 queens? ^{5. 1 from 44 non-king non-queen}

Total $n = 52$
choose 5 $\rightarrow \binom{52}{5}$
$$\frac{\binom{1}{1}\binom{3}{1}\binom{4}{2}\binom{44}{1}}{\binom{52}{5}} = \frac{792}{2598960} = \frac{33}{108290}$$

5. (6 points) An urn contains 4 ^Rred balls, 3 ^Bblue balls, and 3 ^Ggreen balls. Four balls are selected at random **without replacement**. What is the probability that at least one color is not selected?

Total $n = 10$
choose 4 $\rightarrow \binom{10}{4}$
$$1 - \text{All 3 colors selected} = 1 - \frac{126}{\binom{10}{4}} = 1 - \frac{3}{5} = \frac{2}{5}$$

↓
Case #1: 2 R, 1 B, 1 G $\rightarrow \binom{4}{2}\binom{3}{1}\binom{3}{1} = 54$
Case #2: 1 R, 2 B, 1 G $\rightarrow \binom{4}{1}\binom{3}{2}\binom{3}{1} = 36$
Case #3: 1 R, 1 B, 2 G $\rightarrow \binom{4}{1}\binom{3}{1}\binom{3}{2} = 36$

$$W \frac{5}{10} \quad R \frac{3}{10} \quad B \frac{2}{10}$$

6. (6 points) An urn contains 10 balls: 5 are white, 3 are red and 2 are black. Three balls are drawn at random, with replacement, from the urn. What is the probability that all 3 balls are different colors?

Total n: 10

6 Possibilities:

WRB
WBR
RWB
RBW
BWR
BRW

1 order: $\frac{5}{10} \times \frac{3}{10} \times \frac{2}{10} = \frac{30}{1000}$

$6 \times \frac{30}{1000} = \frac{180}{1000} = \frac{9}{50}$

6 orders

7. (9 points) A shelf contains 5 **different** statistics books, 4 **different** algebra books, and 3 **different** calculus books. How many ways can the 12 books be arranged if all statistics books must be together and all algebra books must be together? Assume all books are distinct.

order possibilities for S: 5! $5! \times 4! \times 3! = 120 \times 24 \times 6 = 34560$

order possibilities for A: 4!

order possibilities for all: 5!

↓

S A C1 C2 C3

5 elements

8. (10 points) A fair six-sided die is rolled **five** times. What is the probability of obtaining exactly two 6s and exactly one odd number among the remaining three rolls?

Choose 2 6's: $\binom{5}{2} = 10$

Choose 1 odd #: $\binom{3}{1} = 3$

↓

1, 3, 5

remaining 2 rolls: $\binom{2}{1} \cdot \binom{2}{1} = 4$

↓

can't be 6: 2, 4

3 elements order

$\binom{3}{1} = 3$

$\frac{10 \cdot 3 \cdot 3 \cdot 2 \cdot 2}{6^5} = \frac{360}{776} = \frac{5}{108}$

9. (6 points) A class has 18 first-year students and 12 second-year students. **Five** students are selected at random without replacement. What is the probability that at least 4 of the selected students are second-year students?

Total n: 30

choose 5: $\binom{30}{5}$

Case #1: 4S, 1F

$\frac{\binom{12}{4} \binom{18}{1}}{\binom{30}{5}}$

Case #2: 5S, 0F

$\frac{\binom{12}{5}}{\binom{30}{5}}$

$\geq 4: 4 \text{ or } 5$

$\frac{\binom{12}{4} \binom{18}{1} + \binom{12}{5}}{\binom{30}{5}} = \frac{8910 + 792}{142506} = \frac{77}{1131}$

10. (8 points) Prove or disprove the following set identity: $A \cap (B \setminus C) = (A \cap B) \setminus (A \cap C)$.

$A \cap (B \setminus C)$

Since $B \setminus C = B \cap C^c$

$A \cap (B \setminus C) = A \cap (B \cap C^c)$

$= A \cap B \cap C^c$

$(A \cap B) \setminus (A \cap C) = (A \cap B) \cap (A \cap C)^c$

Since De Morgan's Law: $(A \cap C)^c = A^c \cup C^c$

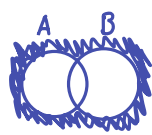
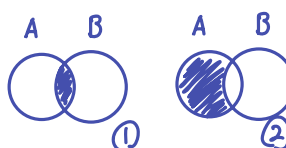
$(A \cap B) \cap (A \cap C)^c = (A \cap B) \cap (A^c \cup C^c)$

$= (A \cap B \cap A^c) \cup (A \cap B \cap C^c)$

Since $A \cap A^c = \emptyset$, $A \cap B \cap A^c = \emptyset$

$= A \cap B \cap C^c$

$A \cap (B \setminus C) = (A \cap B) \setminus (A \cap C)$



11. (9 points) Suppose $P(A) = 0.55$, $P(B) = 0.40$, and $P(A \cup B) = 0.75$. Find $P(A \cap B)$, $P(A \cap B^c)$, and $P(A^c \cap B^c)$.

① $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
 $0.75 = 0.55 + 0.40 - P(A \cap B)$

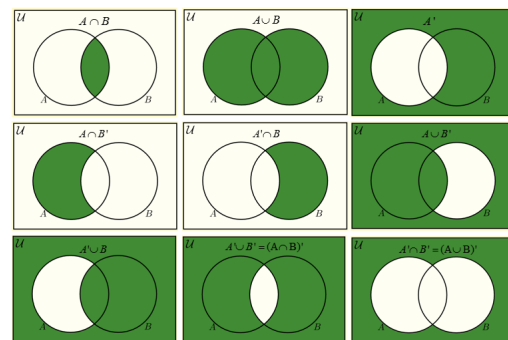
$P(A \cap B) = 0.20$

② $P(A \cap B^c) = P(A) - P(A \cap B)$
 $= 0.55 - 0.20$

$P(A \cap B^c) = 0.35$

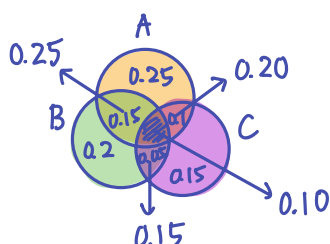
③ $P(A^c \cap B^c) = 1 - P(A \cup B)$
 $= 1 - 0.75$

$P(A^c \cap B^c) = 0.25$



12. (5 points) Suppose $P(A) = 0.60$, $P(B) = 0.50$, $P(C) = 0.40$, $P(A \cap B) = 0.25$, $P(A \cap C) = 0.20$, $P(B \cap C) = 0.15$, and $P(A \cap B \cap C) = 0.10$. Find the probability that at least one event occurs.

At least one event occurs: $P(A \cup B \cup C)$



$\Rightarrow 0.25 + 0.15 + 0.1 + 0.1 + 0.15 + 0.05 + 0.2 = 1$

13. (7 points) A box contains 8 good components and 3 defective components. Four components are selected at random without replacement. What is the probability that **at most** one selected component is defective?

Total $n = 11$

Means either 0 or 1 defective.

Choose 4: $\binom{11}{4}$

Case #1 0 defective: $\binom{8}{4}$

Case #2 1 defective: $\binom{3}{1} \binom{8}{3}$

$\frac{\binom{8}{4} + \binom{3}{1} \binom{8}{3}}{\binom{11}{4}} = \frac{238}{330} = \frac{119}{165}$

14. (9 points) A fair six-sided die is rolled **twice**. What is the probability that

- a six turns up exactly once?
- both numbers are odd?
- the sum of the scores is 4?

a) Case #1: 6, not 6

$\frac{1}{6} \times \frac{5}{6} = \frac{5}{36}$

Case #2: not 6, 6

$\frac{1}{6} \times \frac{5}{6} = \frac{5}{36}$

$2 \times \frac{5}{36} = \frac{5}{18}$

b) Case #1: 1, 1

Case #2: 3, 3

Case #3: 5, 5

Case #4: 1, 3

Case #5: 1, 5

Case #6: 3, 1

Case #7: 3, 5

Case #8: 5, 1

Case #9: 5, 3

$\frac{9}{36} = \frac{1}{4}$

c) Case #1: 1, 3

Case #2: 3, 1

Case #3: 2, 2

$\frac{3}{36} = \frac{1}{12}$